

Simplified RC4 Example

Example

1 Simplified RC4 Example

Lets consider the stream cipher RC4, but instead of the full 256 bytes, we will use 8 x 3-bits. That is, the state vector **S** is 8 x 3-bits. We will operate on 3-bits of plaintext at a time since **S** can take the values 0 to 7, which can be represented as 3 bits.

Assume we use a 4 x 3-bit key of **K** = [1 2 3 6]. And a plaintext **P** = [1 2 2 2]

The first step is to generate the stream.

Initialise the state vector **S** and temporary vector **T**. **S** is initialised so the $S[i] = i$, and **T** is initialised so it is the key **K** (repeated as necessary).

S = [0 1 2 3 4 5 6 7]

T = [1 2 3 6 1 2 3 6]

Now perform the initial permutation on **S**.

```
j = 0;
for i = 0 to 7 do
    j = (j + S[i] + T[i]) mod 8
    Swap(S[i], S[j]);
end
```

For i = 0:

$$\begin{aligned} j &= (0 + 0 + 1) \bmod 8 \\ &= 1 \end{aligned}$$

Swap(S[0],S[1]);

S = [1 0 2 3 4 5 6 7]

For i = 1:

j = 3

Swap(S[1],S[3])

S = [1 3 2 0 4 5 6 7];

For i = 2:

j = 0

Swap(S[2],S[0]);

S = [2 3 1 0 4 5 6 7];

For i = 3:

j = 6;

Swap(S[3],S[6])

S = [2 3 1 6 4 5 0 7];

```
For i = 4:
j = 3
Swap(S[4],S[3])
S = [2 3 1 4 6 5 0 7];
```

```
For i = 5:
j = 2
Swap(S[5],S[2]);
S = [2 3 5 4 6 1 0 7];
```

```
For i = 6:
j = 5;
Swap(S[6],S[4])
S = [2 3 5 4 0 1 6 7];
```

```
For i = 7:
j = 2;
Swap(S[7],S[2])
S = [2 3 7 4 0 1 6 5];
```

Hence, our initial permutation of **S** = [2 3 7 4 0 1 6 5];

Now we generate 3-bits at a time, **k**, that we XOR with each 3-bits of plaintext to produce the ciphertext. The 3-bits **k** is generated by:

```
i, j = 0;
while (true) {
    i = (i + 1) mod 8;
    j = (j + S[i]) mod 8;
    Swap (S[i], S[j]);
    t = (S[i] + S[j]) mod 8;
    k = S[t]; }
```

The first iteration:

```
S = [2 3 7 4 0 1 6 5]
i = (0 + 1) mod 8 = 1
j = (0 + S[1]) mod 8 = 3
Swap(S[1],S[3])
S = [2 4 7 3 0 1 6 5]
t = (S[1] + S[3]) mod 8 = 7
k = S[7] = 5
```

Remember, **P** = [1 2 2 2]

So our first 3-bits of ciphertext is obtained by: **k** XOR **P**
5 XOR 1 = 101 XOR 001 = 100 = 4

The second iteration:

```
S = [2 4 7 3 0 1 6 5]
i = (1 + 1) mod 8 = 2
j = (2 + S[2]) mod 8 = 1
Swap(S[2],S[1])
S = [2 7 4 3 0 1 6 5]
```

$t = (S[2] + S[1]) \bmod 8 = 3$
 $k = S[3] = 3$

Second 3-bits of ciphertext are:
 $3 \text{ XOR } 2 = 011 \text{ XOR } 010 = 001 = 1$

The third iteration:
 $S = [2 \ 7 \ 4 \ 3 \ 0 \ 1 \ 6 \ 5]$
 $i = (2 + 1) \bmod 8 = 3$
 $j = (1 + S[3]) \bmod 8 = 4$
Swap($S[3], S[4]$)
 $S = [2 \ 7 \ 4 \ 0 \ 3 \ 1 \ 6 \ 5]$
 $t = (S[3] + S[4]) \bmod 8 = 3$
 $k = S[3] = 0$

Third 3-bits of ciphertext are:
 $0 \text{ XOR } 2 = 000 \text{ XOR } 010 = 010 = 2$

The final iteration:
 $S = [2 \ 7 \ 4 \ 0 \ 3 \ 1 \ 6 \ 5]$
 $i = (1 + 3) \bmod 8 = 4$
 $j = (4 + S[4]) \bmod 8 = 7$
Swap($S[4], S[7]$)
 $S = [2 \ 7 \ 4 \ 0 \ 5 \ 1 \ 6 \ 3]$
 $t = (S[4] + S[7]) \bmod 8 = 0$
 $k = S[0] = 2$

Last 3-bits of ciphertext are:
 $2 \text{ XOR } 2 = 010 \text{ XOR } 010 = 000 = 0$

So to encrypt the plaintext stream $\mathbf{P} = [1 \ 2 \ 2 \ 2]$ with key $\mathbf{K} = [1 \ 2 \ 3 \ 6]$ using our simplified RC4 stream cipher we get $\mathbf{C} = [4 \ 1 \ 2 \ 0]$.

(or in binary: $\mathbf{P} = 001010010010$, $\mathbf{K} = 001010011110$ and $\mathbf{C} = 100001010000$)